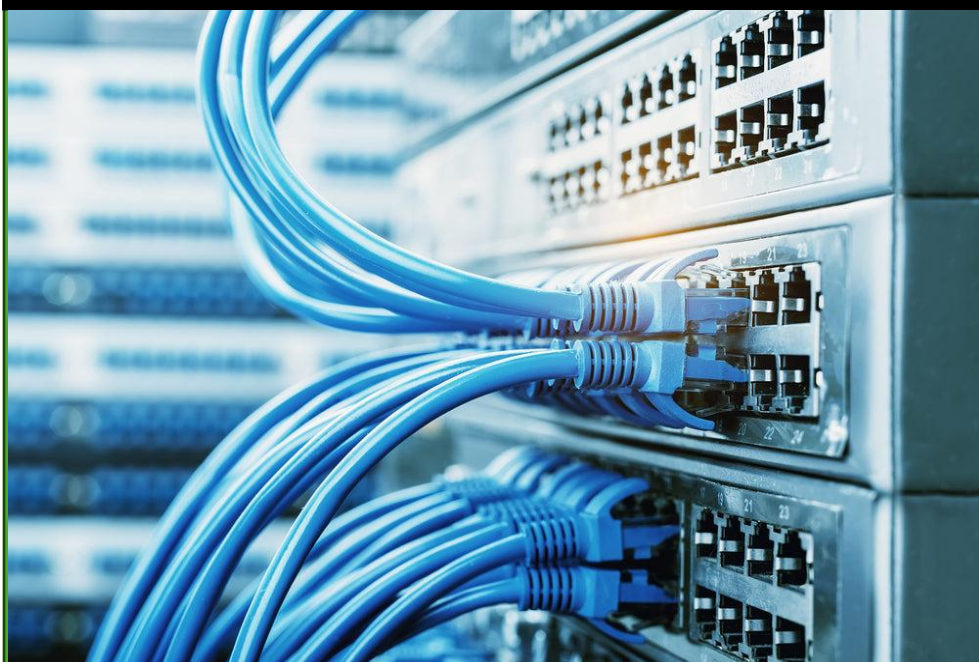




CCNP Lab: Multiprotocol BGP



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Purpose:

The purpose of this lab was to learn the fundamentals of eBGP, IS-IS, OSPFv3, and EIGRP, and how to connect the protocols together in a singular topology by using redistribution commands. We also controlled the routes by changing different BGP attributes such as local preference and weight.

Background:

The origins of BGP go back to the early days of when the internet first began. When the Internet first started out, it was handled by the Exterior Gateway Protocol or EGP. It worked well for the time since it was great with a smaller number of networks, but as the internet began to expand, it was clear that EGP wouldn't be able to sustain the exponential growth of the internet. It lacked the flexibility to enforce routing policies between different administrative domains, and to address this Yakov Rekhter and Kirk Lougheed introduced the very first version of BGP in 1989. It was quickly followed by BGP-2 in 1990, BGP-3 in 1991, and then the current standard mostly used today, BGP-4 which was introduced in 1994. BGP-4 to this day remains the backbone of inter-domain routing across the Internet.

Border Gateway Protocol or BGP is one of the routing protocols which is a foundation for our modern Internet. It is unlike any other protocols that are meant for interior routing such as OSPF, IGRP, or EIGRP. BGP was designed as a way to exchange routing information between entirely separate networks. Each separate network is referred to as an Autonomous System, or AS. Each AS is a collection of connected IP networks and routers under a singular common authority, which is usually an Internet Service Provider, or an ISP. Large enterprises or data centers may also have access to unique AS's that are handed out by the Internet Assigned Numbers Authority or IANA. Each AS is assigned a autonomous system number or ASN for use in BGP, and as such there is a limited quantity mostly given to said enterprises, ISP's and data centers. There are many reserved for practice and public use despite this limitation.

BGP is unique in the way that it finds its paths. While other protocols are distance-vector protocols and consider hop count, and others use a link-state protocol like OSPF to build a complete map of the network, BGP uses a path vector protocol. This means that each route advertisement has a list of all the AS's that the route has traversed, which is called the AS-Path. Through this, the router can detect and avoid routing loops while giving the administrator control over the traffic. For example, a company connected to two AS's can prefer one for outbound traffic based on AS-PATH length, performance considerations, and general preference.

It's also very well known for its stability and scalability. The global internet routing table includes over a million prefixes, and BGP is one of the only routing protocols capable of handling this load. BGP deliberately prioritizes stability unlike protocols which reconverge quickly after a failure, and its slow, incremental updates and timers are

designed to prevent any rapid changes from destabilizing the wider Internet. This design philosophy trades speed for stability to ensure the resilience of global connectivity.

In a router, route redistribution allows a network that uses one routing protocol to route traffic dynamically based on information learned from another routing protocol. Usually, a router will forward data packets based on default/static routes or dynamic routing protocol as it shows in its routing table.

Lab Summary:

In this lab we built a multi-AS eBGP topology which consisted of six routers connected in a line. For every two routers, we had a different protocol. R1 & R2 had OSPFv3, R3 & R4 had EIGRP, and R5 & R6 had IS-IS. The protocols were connected and redistributed by BGP. Finally, we verified connectivity end-to-end and confirmed that the protocols were actually distributed.

Lab Commands:

eBGP Commands:

Command:

```
router bgp AS_number
```

Example:

```
router bgp 65001
```

Purpose:

This starts the BGP routing process for the given autonomous system number. Every router in your network must be configured with the correct AS number for its role.

network *network_address* **mask** *subnet_mask*

Example: *network 192.168.0.0 mask 255.255.255.0*

Purpose: Tells BGP to advertise the specified network into the BGP process. The network must exist in the router's routing table before being advertised.

neighbor *IP_address* **remote-as** *AS_number*

Example: *neighbor 10.0.11.2 remote-as 65002*

Purpose: Defines a BGP neighbor and specifies its autonomous system number. Used to establish a BGP session between routers.

neighbor *IP_address* **next-hop-self**

Example: *neighbor 10.0.15.2 next-hop-self*

Purpose: Configures the router to set itself as the next hop when advertising routes to the specified neighbor. Used in iBGP to ensure next hops are reachable.

OSPFv3 Commands:

Command:

router-id <A.B.C.D>

Example:

router-id 1.1.1.1

Purpose:

Assigns a unique 32-bit router ID for OSPFv3 (used in neighbor relationships).

Command:

ipv6 router ospf <process-id>

Example:

ipv6 router ospf 10

Purpose:

Starts the OSPFv3 process on the router for IPv6 routing.

Command:

interface <interface-name>

Example:

interface GigabitEthernet0/0

Purpose:

Enters interface configuration mode to enable OSPFv3 on a specific interface.

Command:

ipv6 ospf <process-id> area <area-id>

Example:

ipv6 ospf 10 area 0

Purpose:

Enables OSPFv3 on the interface and assigns it to an area.

IS-IS Commands:

Command:

router isis <process-tag>

Example:

router isis CORE

Purpose:

Enables IS-IS routing process.

Command:

net <area-address>.<system-id>.<n-selector>

Example:

net 49.0001.0000.0000.0001.00

Purpose:

Sets the IS-IS network entity title (NET) — unique identifier for the router.

Command:

```
interface <interface-name>
```

Example:

```
interface GigabitEthernet0/0
```

Purpose:

Selects the interface to participate in IS-IS.

Command:

```
ip router isis <process-tag>
```

Example:

```
ip router isis CORE
```

Purpose:

Enables IS-IS on that interface for IPv4.

Command:

```
ipv6 router isis <process-tag>
```

Example:

```
ipv6 router isis CORE
```

Purpose:

Enables IS-IS on that interface for IPv6.

EIGRP Commands:**Command:**

```
router eigrp <asn>
```

Example:

```
router eigrp 100
```

Purpose:

Starts an EIGRP process using the given Autonomous System Number.

Command:

```
network <ip-address> [wildcard-mask]
```

Example:

```
network 192.168.1.0 0.0.0.255
```

Purpose:

Specifies which networks to advertise and on which interfaces to run EIGRP.

Command:

```
router-id <A.B.C.D>
```

Example:

```
router-id 2.2.2.2
```

Purpose:

Sets a unique router ID for EIGRPv6.

Command:

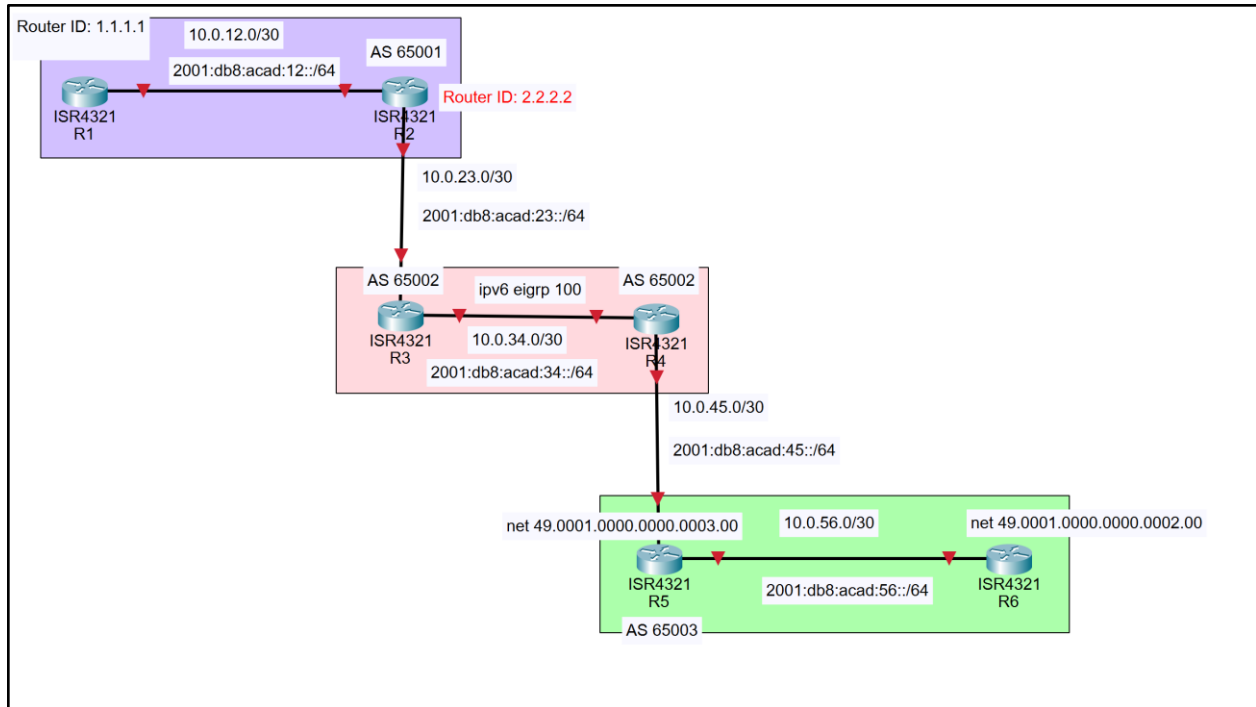
```
ipv6 eigrp <process-id>
```

Example:

```
ipv6 eigrp 100
```

Purpose:

Enables EIGRPv6 on that interface

Network Diagram:**Configs:****R1:**

```
interface Loopback0
ip address 1.1.1.1 255.255.255.255
ipv6 address 2001:DB8:ACAD::1/128
ipv6 enable
ospfv3 1 ipv6 area 0
ospfv3 1 ipv4 area 0

interface GigabitEthernet0/0/0
ip address 10.0.12.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:12::1/64
ipv6 enable
ospfv3 1 ipv4 area 0
```

```
ospfv3 1 ipv6 area 0

interface GigabitEthernet0/0/1
no ip address
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router ospfv3 1
router-id 1.1.1.1

address-family ipv4 unicast
exit-address-family

address-family ipv6 unicast
exit-address-family

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0

control-plane

line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

R2:

```
interface Loopback0

ip address 2.2.2.2 255.255.255.255

ipv6 address 2001:DB8:ACAD::2/128

ipv6 enable
```

```
ospfv3 1 ipv4 area 0
ospfv3 1 ipv6 area 0
```

```
interface GigabitEthernet0/0/0
ip address 10.0.12.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:12::2/64
ipv6 enable
ospfv3 1 ipv4 area 0
ospfv3 1 ipv6 area 0
```

```
interface GigabitEthernet0/0/1
ip address 10.0.23.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:23::1/64
ipv6 enable
```

```
interface Serial0/1/0
no ip address
shutdown
```

```
interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```



```
router odr
```

```
router ospfv3 1
```

```
router-id 2.2.2.2
```

```
address-family ipv4 unicast
```

```
    redistribute connectedv
```

```
    redistribute bgp 65001 metric 10
```

```
exit-address-family
```

```
address-family ipv6 unicast
```

```
    redistribute connected
```

```
    redistribute bgp 65001 metric 10
```

```
exit-address-family
```

```
router bgp 65001
```

```
bgp router-id 2.2.2.2
```

```
bgp log-neighbor-changes
```

```
no bgp default ipv4-unicast
```

```
neighbor 10.0.23.2 remote-as 65002
```

```
neighbor 2001:DB8:ACAD:23::2 remote-as 65002
```

```
address-family ipv4
```

```
    network 10.0.23.0 mask 255.255.255.252
```

```
    redistribute connected
```

```
    redistribute ospfv3 1 metric 10
```

```
    neighbor 10.0.23.2 activate
```

```
    neighbor 10.0.23.2 route-map SET_WEIGHT in
```

```
exit-address-family
```

```
address-family ipv6
```

```
redistribute connected
redistribute ospf 1 metric 10 match internal external 1 external 2
network 2001:DB8:ACAD:23::/64
neighbor 2001:DB8:ACAD:23::2 activate
neighbor 2001:DB8:ACAD:23::2 next-hop-self
exit-address-family
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
route-map SET_WEIGHT permit 10
set weight 250
```

```
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

```
end
```

R3:

```
interface Loopback0
```

```
ip address 3.3.3.3 255.255.255.255
ipv6 address 2001:DB8:ACAD::3/128
ipv6 enable
ipv6 eigrp 100
```

```
interface GigabitEthernet0/0/0
ip address 10.0.34.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:34::1/64
ipv6 enable
ipv6 eigrp 100
```

```
interface GigabitEthernet0/0/1
ip address 10.0.23.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:23::2/64
ipv6 enable
```

```
interface Serial0/1/0
no ip address
shutdown
```

```
interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```

```
router eigrp 100
network 3.3.3.3 0.0.0.0
network 10.0.34.0 0.0.0.3
redistribute bgp 65002 metric 100 1 255 1 1500
redistribute connected
```

```
router eigrp 65002
```

```
router bgp 65002
bgp router-id 3.3.3.3
bgp log-neighbor-changes
neighbor 10.0.23.1 remote-as 65001
neighbor 2001:DB8:ACAD:23::1 remote-as 65001
```

```
address-family ipv4
network 10.0.23.0 mask 255.255.255.252
redistribute connected
redistribute eigrp 100
neighbor 10.0.23.1 activate
neighbor 10.0.23.1 route-map SET_LOCALPREF in
no neighbor 2001:DB8:ACAD:23::1 activate
exit-address-family
```

```
address-family ipv6
redistribute connected
redistribute eigrp 100 metric 10
network 2001:DB8:ACAD:23::/64
neighbor 2001:DB8:ACAD:23::1 activate
neighbor 2001:DB8:ACAD:23::1 next-hop-self
exit-address-family
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ip prefix-list ALL_ROUTES seq 5 permit 0.0.0.0/0 le 32
ipv6 router eigrp 100
eigrp router-id 3.3.3.3
redistribute bgp 65002 metric 10000 100 255 1 1500
redistribute connected
```

```
route-map SET_WEIGHT permit 10
set weight 249
```

```
route-map SET_LOCALPREF permit 10
match ip address prefix-list ALL_ROUTES
set local-preference 150
```

```
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

```
end
```

R3#

R4:

```
interface Loopback0
ip address 4.4.4.4 255.255.255.255
ipv6 address 2001:DB8:ACAD::4/128
ipv6 eigrp 100
```

```
interface GigabitEthernet0/0/0
ip address 10.0.34.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:34::2/64
ipv6 enable
ipv6 eigrp 100
```

```
interface GigabitEthernet0/0/1
ip address 10.0.45.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:45::1/64
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```

```
router eigrp 100
network 4.4.4.4 0.0.0.0
network 10.0.34.0 0.0.0.3
```

```
redistribute bgp 65002 metric 100 1 255 1 1500
redistribute connected
```

```
router bgp 65002
bgp router-id 4.4.4.4
bgp log-neighbor-changes
neighbor 10.0.45.2 remote-as 65003
neighbor 2001:DB8:ACAD:45::2 remote-as 65003
```

```
address-family ipv4
    network 10.0.45.0 mask 255.255.255.252
    redistribute connected
    redistribute eigrp 100 metric 10
    neighbor 10.0.45.2 activate
    neighbor 10.0.45.2 next-hop-self
    no neighbor 2001:DB8:ACAD:45::2 activate
exit-address-family
```

```
address-family ipv6
    redistribute connected
    redistribute eigrp 100 metric 10
    network 2001:DB8:ACAD:45::/64
    neighbor 2001:DB8:ACAD:45::2 activate
    neighbor 2001:DB8:ACAD:45::2 next-hop-self
exit-address-family
```

```
ip http server
ip http authentication local
ip http secure-server
ip forward-protocol nd
ip tftp source-interface GigabitEthernet0
```

```
ipv6 router eigrp 100
eigrp router-id 4.4.4.4
redistribute connected
redistribute bgp 65002 metric 10000 100 255 1 1500
```

```
control-plane
```

```
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
```

```
mgcp profile default
```

```
line con 0
stopbits 1
line aux 0
line vty 0 4
login
transport input ssh
line vty 5 14
login
transport input ssh
```

```
call-home
```

If contact email address in call-home is configured as sch-smart-licensing@cisco.com

the email address configured in Cisco Smart License Portal will be used as contact email address to send SCH notifications.

```
contact-email-addr sch-smart-licensing@cisco.com
```

```
profile "CiscoTAC-1"
```



```
active
destination transport-method http

end
```

R5:

```
interface Loopback0
ip address 5.5.5.5 255.255.255.255
ip router isis CORE
ipv6 address 2001:DB8::5/128
ipv6 router isis CORE

interface GigabitEthernet0/0/0
ip address 10.0.56.1 255.255.255.252
ip router isis CORE
negotiation auto
ipv6 address 2001:DB8:ACAD:56::1/64
ipv6 enable
ipv6 router isis CORE

interface GigabitEthernet0/0/1
ip address 10.0.45.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:45::2/64

interface GigabitEthernet0/1/0
negotiation auto

interface GigabitEthernet0/1/1
```

negotiation auto

interface GigabitEthernet0

vrf forwarding Mgmt-intf

no ip address

shutdown

negotiation auto

interface Vlan1

no ip address

shutdown

router isis CORE

net 49.0001.0000.0000.0003.00

is-type level-2-only

redistribute connected metric 10

redistribute bgp 65003 metric 10

address-family ipv6

redistribute connected metric 10

redistribute bgp 65003 metric 10

exit-address-family

router bgp 65003

bgp log-neighbor-changes

bgp default local-preference 200

neighbor 10.0.45.1 remote-as 65002

neighbor 2001:DB8:ACAD:45::1 remote-as 65002

address-family ipv4

network 10.0.45.0 mask 255.255.255.252

```
redistribute connected metric 10
redistribute isis CORE level-2 metric 10
neighbor 10.0.45.1 activate
no neighbor 2001:DB8:ACAD:45::1 activate
exit-address-family
```

```
address-family ipv6
redistribute connected metric 10
redistribute isis CORE metric 10 level-2
network 2001:DB8:ACAD:45::/64
neighbor 2001:DB8:ACAD:45::1 activate
neighbor 2001:DB8:ACAD:45::1 next-hop-self
exit-address-family
```

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
ip prefix-list net4 seq 5 permit 6.6.6.6/32
```

```
route-map PREF permit 10
match ip address prefix-list net4
set local-preference 300
```

```
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
```

```
line vty 0 4
login
transport input ssh
line vty 5 14
login
transport input ssh

end
```

R6:

```
interface Loopback0
ip address 6.6.6.6 255.255.255.255
ip router isis CORE
ipv6 address 2001:DB8:ACAD::6/128
ipv6 enable
ipv6 router isis CORE

interface GigabitEthernet0/0/0
ip address 10.0.56.2 255.255.255.252
ip router isis CORE
negotiation auto
ipv6 address 2001:DB8:ACAD:56::2/64
ipv6 enable
ipv6 router isis CORE

interface GigabitEthernet0/0/1
no ip address
negotiation auto

interface Serial0/1/0
```

```
no ip address
shutdown
```

```
interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```

```
router isis CORE
net 49.0001.0000.0000.0002.00
is-type level-2-only
metric-style narrow
```

```
router bgp 65003
bgp log-neighbor-changes
bgp default local-preference 200
neighbor 10.0.56.1 remote-as 65003
```

```
address-family ipv4
    neighbor 10.0.56.1 activate
exit-address-family
```

```
ip forward-protocol nd
no ip http server
ip http authentication local
no ip http secure-server
```

```
ip tftp source-interface GigabitEthernet0
```

```
control-plane
```

```
line con 0
```

```
transport input none
```

```
stopbits 1
```

```
line aux 0
```

```
stopbits 1
```

```
line vty 0 4
```

```
Login
```

Show Commands/Connectivity Tests:

Show ip route (all routers), show ospf & ipv6 (r1-r2), show ipv6 route (all routers), show is-is (r5-r6), show eigrp (r3-4), show traceroute (r1, r6), show weights (r3)

R1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets

```

C      1.1.1.1 is directly connected, Loopback0
      2.0.0.0/32 is subnetted, 1 subnets
O      2.2.2.2 [110/1] via 10.0.12.2, 00:07:59, GigabitEthernet0/0/0
      3.0.0.0/32 is subnetted, 1 subnets
O E2   3.3.3.3 [110/10] via 10.0.12.2, 00:07:59, GigabitEthernet0/0/0
      4.0.0.0/32 is subnetted, 1 subnets
O E2   4.4.4.4 [110/10] via 10.0.12.2, 00:07:02, GigabitEthernet0/0/0
      5.0.0.0/32 is subnetted, 1 subnets
O E2   5.5.5.5 [110/10] via 10.0.12.2, 00:07:02, GigabitEthernet0/0/0
      6.0.0.0/32 is subnetted, 1 subnets
O E2   6.6.6.6 [110/10] via 10.0.12.2, 00:06:31, GigabitEthernet0/0/0
      10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C      10.0.12.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.12.1/32 is directly connected, GigabitEthernet0/0/0
O E2   10.0.23.0/30 [110/10] via 10.0.12.2, 00:07:59, GigabitEthernet0/0/0
O E2   10.0.34.0/30 [110/10] via 10.0.12.2, 00:07:33, GigabitEthernet0/0/0
O E2   10.0.45.0/30 [110/10] via 10.0.12.2, 00:07:02, GigabitEthernet0/0/0
O E2   10.0.56.0/30 [110/10] via 10.0.12.2, 00:07:02, GigabitEthernet0/0/0

```

R1#show ipv6 route

IPv6 Routing Table - default - 13 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```
OE2 2001:DB8::5/128 [110/10]
```

```
      via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
```

```
LC 2001:DB8:ACAD::1/128 [0/0]
```

```

    via Loopback0, receive
O  2001:DB8:ACAD::2/128 [110/1]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD::3/128 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD::4/128 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD::6/128 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
C  2001:DB8:ACAD:12::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L  2001:DB8:ACAD:12::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
OE2 2001:DB8:ACAD:23::/64 [110/20]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:34::/64 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:45::/64 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:56::/64 [110/10]
    via FE80::227:90FF:FED5:FA40, GigabitEthernet0/0/0
L  FF00::/8 [0/0]
    via Null0, receive
R1#

```

R1#show ospfv3 neighbor

```

OSPFv3 1 address-family ipv4 (router-id 1.1.1.1)

```

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
-------------	-----	-------	-----------	--------------	-----------


```
2.2.2.2          1    FULL/DR          00:00:31    6
GigabitEthernet0/0/0
```

OSPFv3 1 address-family ipv6 (router-id 1.1.1.1)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
2.2.2.2	1	FULL/DR	00:00:39	6	
GigabitEthernet0/0/0					

R1#traceroute ip 10.0.56.2

Type escape sequence to abort.

Tracing the route to 10.0.56.2

VRF info: (vrf in name/id, vrf out name/id)

```
 1 10.0.12.2 1 msec 1 msec 1 msec
 2 10.0.23.2 1 msec 1 msec 1 msec
 3 10.0.34.2 1 msec 1 msec 1 msec
 4 10.0.45.2 1 msec 1 msec 1 msec
 5 10.0.56.2 2 msec 2 msec *
```

R1#traceroute ipv6 2001:db8:acad:56::2

Type escape sequence to abort.

Tracing the route to 2001:DB8:ACAD:56::2

```
 1 2001:DB8:ACAD:12::2 6 msec 1 msec 1 msec
 2 2001:DB8:ACAD:23::2 1 msec 2 msec 1 msec
 3 2001:DB8:ACAD:34::2 1 msec 2 msec 1 msec
 4 2001:DB8:ACAD:45::2 1 msec 1 msec 1 msec
 5 2001:DB8:ACAD:56::2 1 msec 2 msec 1 msec
```

R2#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static
 route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

1.0.0.0/32 is subnetted, 1 subnets
O      1.1.1.1 [110/1] via 10.0.12.1, 00:15:28, GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
C      2.2.2.2 is directly connected, Loopback0
3.0.0.0/32 is subnetted, 1 subnets
B      3.3.3.3 [20/0] via 10.0.23.2, 00:15:31
4.0.0.0/32 is subnetted, 1 subnets
B      4.4.4.4 [20/130816] via 10.0.23.2, 00:14:31
5.0.0.0/32 is subnetted, 1 subnets
B      5.5.5.5 [20/25600512] via 10.0.23.2, 00:14:31
  
```

R2#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

```

B    2001:DB8::5/128 [20/10]
    via FE80::227:90FF:FED5:FAD1, GigabitEthernet0/0/1
O    2001:DB8:ACAD::1/128 [110/1]
    via FE80::CE7F:76FF:FE6A:83D0, GigabitEthernet0/0/0
LC   2001:DB8:ACAD::2/128 [0/0]
    via Loopback0, receive
B    2001:DB8:ACAD::3/128 [20/0]
    via FE80::227:90FF:FED5:FAD1, GigabitEthernet0/0/1
B    2001:DB8:ACAD::4/128 [20/10]
    via FE80::227:90FF:FED5:FAD1, GigabitEthernet0/0/1
B    2001:DB8:ACAD::6/128 [20/10]
    via FE80::227:90FF:FED5:FAD1, GigabitEthernet0/0/1
C    2001:DB8:ACAD:12::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:12::2/128 [0/0]
    via GigabitEthernet0/0/0, receive

```

R2#show ospfv3 neighbor

OSPFv3 1 address-family ipv4 (router-id 2.2.2.2)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/BDR	00:00:34	6	
GigabitEthernet0/0/0					

OSPFv3 1 address-family ipv6 (router-id 2.2.2.2)

Neighbor ID	Pri	State	Dead Time	Interface ID	Interface
1.1.1.1	1	FULL/BDR	00:00:31	6	
GigabitEthernet0/0/0					

R2#

R3#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets

B 1.1.1.1 [20/10] via 10.0.23.1, 00:19:07

2.0.0.0/32 is subnetted, 1 subnets

B 2.2.2.2 [20/0] via 10.0.23.1, 00:19:37

3.0.0.0/32 is subnetted, 1 subnets

C 3.3.3.3 is directly connected, Loopback0

4.0.0.0/32 is subnetted, 1 subnets

D 4.4.4.4 [90/130816] via 10.0.34.2, 00:19:06, GigabitEthernet0/0/0

5.0.0.0/32 is subnetted, 1 subnets

D EX 5.5.5.5 [170/25600512] via 10.0.34.2, 00:18:58, GigabitEthernet0/0/0

6.0.0.0/32 is subnetted, 1 subnets

D EX 6.6.6.6 [170/25600512] via 10.0.34.2, 00:18:28, GigabitEthernet0/0/0

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks

B 10.0.12.0/30 [20/0] via 10.0.23.1, 00:19:37

C 10.0.23.0/30 is directly connected, GigabitEthernet0/0/1

L 10.0.23.2/32 is directly connected, GigabitEthernet0/0/1

C 10.0.34.0/30 is directly connected, GigabitEthernet0/0/0

```
L      10.0.34.1/32 is directly connected, GigabitEthernet0/0/0
D EX   10.0.45.0/30 [170/3072] via 10.0.34.2, 00:19:01,
GigabitEthernet0/0/0
D EX   10.0.56.0/30
      [170/25600512] via 10.0.34.2, 00:18:58, GigabitEthernet0/0/0
R3#
```

R3#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, la - LISP alt

lr - LISP site-registrations, ld - LISP dyn-eid, la - LISP away

le - LISP extranet-policy, a - Application

```
EX 2001:DB8::5/128 [170/281856], tag 65003
    via FE80::DEF7:19FF:FE78:4890, GigabitEthernet0/0/0
B   2001:DB8:ACAD::1/128 [20/10]
    via FE80::227:90FF:FED5:FA41, GigabitEthernet0/0/1
B   2001:DB8:ACAD::2/128 [20/0]
    via FE80::227:90FF:FED5:FA41, GigabitEthernet0/0/1
LC  2001:DB8:ACAD::3/128 [0/0]
    via Loopback0, receive
D   2001:DB8:ACAD::4/128 [90/130816]
    via FE80::DEF7:19FF:FE78:4890, GigabitEthernet0/0/0
EX  2001:DB8:ACAD::6/128 [170/281856], tag 65003
    via FE80::DEF7:19FF:FE78:4890, GigabitEthernet0/0/0
B   2001:DB8:ACAD:12::/64 [20/0]
    via FE80::227:90FF:FED5:FA41, GigabitEthernet0/0/1
```

```

C    2001:DB8:ACAD:23::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:DB8:ACAD:23::2/128 [0/0]
    via GigabitEthernet0/0/1, receive
C    2001:DB8:ACAD:34::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:34::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
EX   2001:DB8:ACAD:45::/64 [170/3072]
    via FE80::DEF7:19FF:FE78:4890, GigabitEthernet0/0/0
EX   2001:DB8:ACAD:56::/64 [170/281856], tag 65003
    via FE80::DEF7:19FF:FE78:4890, GigabitEthernet0/0/0
L    FF00::/8 [0/0]
    via Null0, receive
R3#

```

R3#show ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold Uptime	SRTT	RTO
Q	Seq		(sec)	(ms)	
Cnt	Num				
0	10.0.34.2	Gi0/0/0	14 00:19:28	2	100
0	6				

EIGRP-IPv4 Neighbors for AS(65002)

R3#show ip bgp

BGP table version is 15, local router ID is 3.3.3.3

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	1.1.1.1/32	10.0.23.1	10	150	249	65001 ?
*>	2.2.2.2/32	10.0.23.1	0	150	249	65001 ?
*>	3.3.3.3/32	0.0.0.0	0		32768	?
*>	4.4.4.4/32	10.0.34.2	130816		32768	?
*>	5.5.5.5/32	10.0.34.2	25600512		32768	?
*>	6.6.6.6/32	10.0.34.2	25600512		32768	?
*>	10.0.12.0/30	10.0.23.1	0	150	249	65001 ?
*	10.0.23.0/30	10.0.23.1	0	150	249	65001 i
*>		0.0.0.0	0		32768	i
*>	10.0.34.0/30	0.0.0.0	0		32768	?
*>	10.0.45.0/30	10.0.34.2	3072		32768	?
*>	10.0.56.0/30	10.0.34.2	25600512		32768	?

R4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PfR

& - replicated local route overrides by connected

Gateway of last resort is not set

```
1.0.0.0/32 is subnetted, 1 subnets
D EX    1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53, GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
D EX    2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
3.0.0.0/32 is subnetted, 1 subnets
D        3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
4.0.0.0/32 is subnetted, 1 subnets
C        4.4.4.4 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
B        5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
6.0.0.0/32 is subnetted, 1 subnets
B        6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX    10.0.12.0/30
[170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX    10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C        10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L        10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C        10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L        10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B        10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45
```

R4#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE - Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application, m - OMP

B 2001:DB8::5/128 [20/10], tag 65003

via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

EX 2001:DB8:ACAD::1/128 [170/281856]

via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

EX 2001:DB8:ACAD::2/128 [170/281856]

via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

D 2001:DB8:ACAD::3/128 [90/130816]

via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

LC 2001:DB8:ACAD::4/128 [0/0]

via Loopback0, receive

B 2001:DB8:ACAD::6/128 [20/10], tag 65003

via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:12::/64 [170/281856]

via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

EX 2001:DB8:ACAD:23::/64 [170/3072]

via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

C 2001:DB8:ACAD:34::/64 [0/0]

via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:34::2/128 [0/0]

via GigabitEthernet0/0/0, receive

C 2001:DB8:ACAD:45::/64 [0/0]

via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:45::1/128 [0/0]

via GigabitEthernet0/0/1, receive

B 2001:DB8:ACAD:56::/64 [20/10], tag 65003

via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

L FF00::/8 [0/0]

via Null0, receive

R4#show ip bgp

BGP table version is 12, local router ID is 4.4.4.4

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

t secondary path, L long-lived-stale,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	1.1.1.1/32	10.0.34.1	10		32768	?
*>	2.2.2.2/32	10.0.34.1	10		32768	?
*>	3.3.3.3/32	10.0.34.1	10		32768	?
*>	4.4.4.4/32	0.0.0.0	0		32768	?
*>	5.5.5.5/32	10.0.45.2	10		0 65003	?
*>	6.6.6.6/32	10.0.45.2	10		0 65003	?
*>	10.0.12.0/30	10.0.34.1	10		32768	?
*>	10.0.23.0/30	10.0.34.1	10		32768	?
*>	10.0.34.0/30	0.0.0.0	0		32768	?
*	10.0.45.0/30	10.0.45.2	0		0 65003	i
*>		0.0.0.0	0		32768	i
*>	10.0.56.0/30	10.0.45.2	10		0 65003	?

R4#show bgp ipv6

% Command accepted but obsolete, unreleased or unsupported; see documentation.

BGP table version is 12, local router ID is 4.4.4.4

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
x best-external, a additional-path, c RIB-compressed,
t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2001:DB8::5/128	2001:DB8:ACAD:45::2				
10		0	65003			?
*>	2001:DB8:ACAD::1/128					
	FE80::227:90FF:FED5:FAD0					
10		32768				?
*>	2001:DB8:ACAD::2/128					
	FE80::227:90FF:FED5:FAD0					
10		32768				?
*>	2001:DB8:ACAD::3/128					
	FE80::227:90FF:FED5:FAD0					
10		32768				?
*>	2001:DB8:ACAD::4/128					
::		0				32768 ?
*>	2001:DB8:ACAD::6/128					
	2001:DB8:ACAD:45::2					
10		0	65003			?
*>	2001:DB8:ACAD:12::/64					
	FE80::227:90FF:FED5:FAD0					
10		32768				?
*>	2001:DB8:ACAD:23::/64					
	FE80::227:90FF:FED5:FAD0					
10		32768				?

*> 2001:DB8:ACAD:34::/64

:: 0 32768 ?

* 2001:DB8:ACAD:45::/64

2001:DB8:ACAD:45::2

10 0 65003 i

*> :: 0 32768 i

*> 2001:DB8:ACAD:56::/64

2001:DB8:ACAD:45::2

10 0 65003 ?

R4#show ip eigrp neighbors

EIGRP-IPv4 Neighbors for AS(100)

H	Address	Interface	Hold	Uptime	SRTT	RTO
Q	Seq					
(sec)	(ms)	Cnt Num				
0	10.0.34.1	Gi0/0/0	12	00:24:41	814	4884 0
7						

R4#show ipv6 eigrp neighbors

EIGRP-IPv6 Neighbors for AS(100)

H	Address	Interface	Hold	Uptime	SRTT	RTO
Q	Seq					
(sec)	(ms)	Cnt Num				
0	Link-local address:	Gi0/0/0	10	00:24:48	652	3912 0
8						

FE80::227:90FF:FED5:FAD0

R4#show ip eigrp topology

EIGRP-IPv4 Topology Table for AS(100)/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,

r - reply Status, s - sia Status

P 10.0.34.0/30, 1 successors, FD is 2816

via Connected, GigabitEthernet0/0/0

P 4.4.4.4/32, 1 successors, FD is 128256

```
via Connected, Loopback0
P 10.0.56.0/30, 1 successors, FD is 25600256, tag is 65003
    via Redistributed (25600256/0)
P 5.5.5.5/32, 1 successors, FD is 25600256, tag is 65003
    via Redistributed (25600256/0)
P 2.2.2.2/32, 1 successors, FD is 25600512, tag is 65001
    via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0
P 6.6.6.6/32, 1 successors, FD is 25600256, tag is 65003
    via Redistributed (25600256/0)
P 10.0.45.0/30, 1 successors, FD is 2816
    via Rconnected (2816/0)
P 10.0.23.0/30, 1 successors, FD is 3072
    via 10.0.34.1 (3072/2816), GigabitEthernet0/0/0
P 3.3.3.3/32, 1 successors, FD is 130816
    via 10.0.34.1 (130816/128256), GigabitEthernet0/0/0
P 1.1.1.1/32, 1 successors, FD is 25600512, tag is 65001
    via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0
P 10.0.12.0/30, 1 successors, FD is 25600512, tag is 65001
    via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0
```

```
R4#show ipv6 eigrp topology
```

```
EIGRP-IPv6 Topology Table for AS(100)/ID(4.4.4.4)
```

```
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status
```

```
P 2001:DB8::5/128, 1 successors, FD is 281600, tag is 65003
    via Redistributed (281600/0)
P 2001:DB8:ACAD:45::/64, 1 successors, FD is 2816
    via Rconnected (2816/0)
P 2001:DB8:ACAD:56::/64, 1 successors, FD is 281600, tag is 65003
    via Redistributed (281600/0)
```

```

P 2001:DB8:ACAD::2/128, 1 successors, FD is 281856
    via FE80::227:90FF:FED5:FAD0 (281856/281600), GigabitEthernet0/0/0
P 2001:DB8:ACAD:12::/64, 1 successors, FD is 281856
    via FE80::227:90FF:FED5:FAD0 (281856/281600), GigabitEthernet0/0/0
P 2001:DB8:ACAD:34::/64, 1 successors, FD is 2816
    via Connected, GigabitEthernet0/0/0
P 2001:DB8:ACAD::1/128, 1 successors, FD is 281856
    via FE80::227:90FF:FED5:FAD0 (281856/281600), GigabitEthernet0/0/0
P 2001:DB8:ACAD::3/128, 1 successors, FD is 130816
    via FE80::227:90FF:FED5:FAD0 (130816/128256), GigabitEthernet0/0/0
P 2001:DB8:ACAD::4/128, 1 successors, FD is 128256
    via Connected, Loopback0
P 2001:DB8:ACAD::6/128, 1 successors, FD is 281600, tag is 65003
    via Redistributed (281600/0)
P 2001:DB8:ACAD:23::/64, 1 successors, FD is 3072
    via FE80::227:90FF:FED5:FAD0 (3072/2816), GigabitEthernet0/0/0

```

R5:

```
R5#show ip route
```

```
*Nov 17 17:55:54.649: %SYS-5-CONFIG_I: Configured from console by console
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
ia - IS-IS inter area, * - candidate default, U - per-user static
```

```
route
```

```
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
```

```
a - application route
```

```
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```

1.0.0.0/32 is subnetted, 1 subnets
B      1.1.1.1 [20/10] via 10.0.45.1, 00:16:43
2.0.0.0/32 is subnetted, 1 subnets
B      2.2.2.2 [20/10] via 10.0.45.1, 00:16:43
3.0.0.0/32 is subnetted, 1 subnets
B      3.3.3.3 [20/10] via 10.0.45.1, 00:16:43
4.0.0.0/32 is subnetted, 1 subnets
B      4.4.4.4 [20/0] via 10.0.45.1, 00:16:43
5.0.0.0/32 is subnetted, 1 subnets
C      5.5.5.5 is directly connected, Loopback0
6.0.0.0/32 is subnetted, 1 subnets
i L2   6.6.6.6 [115/20] via 10.0.56.2, 00:16:32, GigabitEthernet0/0/0
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
B      10.0.12.0/30 [20/10] via 10.0.45.1, 00:16:43
B      10.0.23.0/30 [20/10] via 10.0.45.1, 00:16:43
B      10.0.34.0/30 [20/0] via 10.0.45.1, 00:16:43
C      10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L      10.0.45.2/32 is directly connected, GigabitEthernet0/0/1
C      10.0.56.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.56.1/32 is directly connected, GigabitEthernet0/0/0

```

R5#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

LC 2001:DB8::5/128 [0/0]

via Loopback0, receive

```

B    2001:DB8:ACAD::1/128 [20/10]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::2/128 [20/10]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::3/128 [20/10]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::4/128 [20/0]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
I2  2001:DB8:ACAD::6/128 [115/20]
      via FE80::521C:B0FF:FE42:E970, GigabitEthernet0/0/0
B    2001:DB8:ACAD:12::/64 [20/10]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD:23::/64 [20/10]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD:34::/64 [20/0]
      via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
C    2001:DB8:ACAD:45::/64 [0/0]
      via GigabitEthernet0/0/1, directly connected
L    2001:DB8:ACAD:45::2/128 [0/0]
      via GigabitEthernet0/0/1, receive
C    2001:DB8:ACAD:56::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:56::1/128 [0/0]
      via GigabitEthernet0/0/0, receive
L    FF00::/8 [0/0]
      via Null0, receive

```

```
R5#show isis * rib
```

IPv4 local RIB for IS-IS process CORE

IPV4 unicast topology base (TID 0, TOPOID 0x0) =====

6.6.6.6/32

[115/L2/20] via 10.0.56.2(GigabitEthernet0/0/0), from 6.6.6.6, tag 0,
LSP[2/2]

(installed)

10.0.56.0/30

[115/L2/20] via 10.0.56.2(GigabitEthernet0/0/0), from 6.6.6.6, tag 0,
LSP[2/2]

IS-IS IPv6 process CORE, local RIB

* 2001:DB8:ACAD::6/128

via FE80::521C:B0FF:FE42:E970/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/2]

2001:DB8:ACAD:56::/64

via FE80::521C:B0FF:FE42:E970/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/2]

R5# show isis * topology

Tag CORE:

IS-IS TID 0 paths to level-2 routers

System Id	Metric	Next-Hop	Interface	SNPA
R6 501c.b042.e970	10	R6	Gi0/0/0	
R5	--			

R5#show isis neighbors

Tag CORE:

System Id	Type	Interface	IP Address	State	Holdtime	Circuit Id
R6	L2	Gi0/0/0	10.0.56.2	UP	28	R5.01

R5#show ip bgp

BGP table version is 12, local router ID is 5.5.5.5

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 1.1.1.1/32	10.0.45.1	10		0	65002 ?
*> 2.2.2.2/32	10.0.45.1	10		0	65002 ?
*> 3.3.3.3/32	10.0.45.1	10		0	65002 ?
*> 4.4.4.4/32	10.0.45.1	0		0	65002 ?
*> 5.5.5.5/32	0.0.0.0	10		32768	?
*> 6.6.6.6/32	10.0.56.2	10		32768	?
*> 10.0.12.0/30	10.0.45.1	10		0	65002 ?
*> 10.0.23.0/30	10.0.45.1	10		0	65002 ?
*> 10.0.34.0/30	10.0.45.1	0		0	65002 ?
* 10.0.45.0/30	10.0.45.1	0		0	65002 iR4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets
D EX 1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53, GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
D EX 2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
3.0.0.0/32 is subnetted, 1 subnets
D 3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
4.0.0.0/32 is subnetted, 1 subnets
C 4.4.4.4 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
B 5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
6.0.0.0/32 is subnetted, 1 subnets
B 6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX 10.0.12.0/30
[170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX 10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C 10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L 10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C 10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L 10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B 10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45

R4#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
 I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
 EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -
 Destination
 NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
 OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
 a - Application, m - OMP

B 2001:DB8::5/128 [20/10], tag 65003
 via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

EX 2001:DB8:ACAD::1/128 [170/281856]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

EX 2001:DB8:ACAD::2/128 [170/281856]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

D 2001:DB8:ACAD::3/128 [90/130816]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

LC 2001:DB8:ACAD::4/128 [0/0]
 via Loopback0, receive

B 2001:DB8:ACAD::6/128 [20/10], tag 65003
 via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:12::/64 [170/281856]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

EX 2001:DB8:ACAD:23::/64 [170/3072]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

C 2001:DB8:ACAD:34::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:34::2/128 [0/0]
 via GigabitEthernet0/0/0, receive

C 2001:DB8:ACAD:45::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:45::1/128 [0/0]
 via GigabitEthernet0/0/1, receive

B 2001:DB8:ACAD:56::/64 [20/10], tag 65003
via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

L FF00::/8 [0/0]
via Null0, receive

```
*>          0.0.0.0          0          32768 i
*> 10.0.56.0/30  0.0.0.0          10          32768 ?
```

R5#show bgp ipv6 unicast

BGP table version is 20, local router ID is 5.5.5.5

Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-
Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 2001:DB8::5/128	::	10		32768	?
*> 2001:DB8:ACAD::1/128					
2001:DB8:ACAD:45::1					
10	0	65002			?
*> 2001:DB8:ACAD::2/128					
2001:DB8:ACAD:45::1					
10	0	65002			?
*> 2001:DB8:ACAD::3/128					
2001:DB8:ACAD:45::1					
10	0	65002			?
*> 2001:DB8:ACAD::4/128					
2001:DB8:ACAD:45::1					
0	0	65002			?
*> 2001:DB8:ACAD::6/128					

Network	Next Hop	Metric	LocPrf	Weight	Path
---------	----------	--------	--------	--------	------

FE80::521C:B0FF:FE42:E970

10 32768 ?

*> 2001:DB8:ACAD:12::/64

2001:DB8:ACAD:45::1

10 0 65002 ?

*> 2001:DB8:ACAD:23::/64

2001:DB8:ACAD:45::1

10 0 65002 ?

*> 2001:DB8:ACAD:34::/64

2001:DB8:ACAD:45::1

0 0 65002 ?

* 2001:DB8:ACAD:45::/64

2001:DB8:ACAD:45::1

0 0 65002 i

*> :: 10 32768 i

*> 2001:DB8:ACAD:56::/64

:: 10 32768 ?

R6:

R6(config)#do traceroute 1.1.1.1

Type escape sequence to abort.

Tracing the route to 1.1.1.1

VRF info: (vrf in name/id, vrf out name/id)

1 10.0.56.1 1 msec 1 msec 0 msec

2 10.0.45.1 1 msec 1 msec 0 msec

3 10.0.34.1 1 msec 2 msec 1 msec

4 10.0.23.1 1 msec 2 msec 1 msec

5 10.0.12.1 2 msec 2 msec *

R6(config)#do traceroute 2001:db8:acad::1

Type escape sequence to abort.

Tracing the route to 2001:DB8:ACAD::1

```

1 2001:DB8:ACAD:56::1 6 msec 1 msec 0 msec
2 2001:DB8:ACAD:45::1 14 msec 1 msec 0 msec
3 2001:DB8:ACAD:34::1 1 msec 2 msec 1 msec
4 2001:DB8:ACAD:23::1 1 msec 9 msec 2 msec
5 2001:DB8:ACAD:12::1 1 msec 2 msec 2 msec

```

R6#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSR4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static

route

H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR
& - replicated local route overrides by connected

Gateway of last resort is not set

```

1.0.0.0/32 is subnetted, 1 subnets
D EX    1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53, GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
D EX    2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
3.0.0.0/32 is subnetted, 1 subnets
D       3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0

```

```

    4.0.0.0/32 is subnetted, 1 subnets
C      4.4.4.4 is directly connected, Loopback0
    5.0.0.0/32 is subnetted, 1 subnets
B      5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
    6.0.0.0/32 is subnetted, 1 subnets
B      6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX    10.0.12.0/30
[170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX    10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C      10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C      10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L      10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B      10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45
R4#show ipv6 route
IPv6 Routing Table - default - 14 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
        B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
        I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
        EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -
Destination
        NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
        OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        a - Application, m - OMP
B  2001:DB8::5/128 [20/10], tag 65003
    via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX 2001:DB8:ACAD::1/128 [170/281856]
    via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX 2001:DB8:ACAD::2/128 [170/281856]
    via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

```


D 2001:DB8:ACAD::3/128 [90/130816]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

LC 2001:DB8:ACAD::4/128 [0/0]
 via Loopback0, receive

B 2001:DB8:ACAD::6/128 [20/10], tag 65003
 via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

EX 2001:DB8:ACAD:12::/64 [170/281856]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

EX 2001:DB8:ACAD:23::/64 [170/3072]
 via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0

C 2001:DB8:ACAD:34::/64 [0/0]
 via GigabitEthernet0/0/0, directly connected

L 2001:DB8:ACAD:34::2/128 [0/0]
 via GigabitEthernet0/0/0, receive

C 2001:DB8:ACAD:45::/64 [0/0]
 via GigabitEthernet0/0/1, directly connected

L 2001:DB8:ACAD:45::1/128 [0/0]
 via GigabitEthernet0/0/1, receive

B 2001:DB8:ACAD:56::/64 [20/10], tag 65003
 via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

L FF00::/8 [0/0]
 via Null0, receive

A external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static
 route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

1.0.0.0/32 is subnetted, 1 subnets
i L2    1.1.1.1 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
i L2    2.2.2.2 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
3.0.0.0/32 is subnetted, 1 subnets
i L2    3.3.3.3 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
4.0.0.0/32 is subnetted, 1 subnets
i L2    4.4.4.4 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
5.0.0.0/32 is subnetted, 1 subnets
i L2    5.5.5.5 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
6.0.0.0/32 is subnetted, 1 subnets
C        6.6.6.6 is directly connected, Loopback0
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
i L2    10.0.12.0/30 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
i L2    10.0.23.0/30 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
i L2    10.0.34.0/30 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
i L2    10.0.45.0/30 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
C        10.0.56.0/30 is directly connected, GigabitEthernet0/0/0
L        10.0.56.2/32 is directly connected, GigabitEthernet0/0/0

```

R6#show ipv6 route

IPv6 Routing Table - default - 13 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

I2 2001:DB8::5/128 [115/20]

```

        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD::1/128 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD::2/128 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD::3/128 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD::4/128 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
LC 2001:DB8:ACAD::6/128 [0/0]
        via Loopback0, receive
I2 2001:DB8:ACAD:12::/64 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD:23::/64 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD:34::/64 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2 2001:DB8:ACAD:45::/64 [115/20]
        via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
C   2001:DB8:ACAD:56::/64 [0/0]
        via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:56::2/128 [0/0]
        via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
        via Null0, receive
R6#show isis * rib

```

IPv4 local RIB for IS-IS process CORE

IPv4 unicast topology base (TID 0, TOPOID 0x0) =====

1.1.1.1/32

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

2.2.2.2/32

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

3.3.3.3/32

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

4.4.4.4/32

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

5.5.5.5/32

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

10.0.12.0/30

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

10.0.23.0/30

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

10.0.34.0/30

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

10.0.45.0/30

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

(installed)

10.0.56.0/30

[115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]

IS-IS IPv6 process CORE, local RIB

* 2001:DB8::5/128

via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/4]

* 2001:DB8:ACAD::1/128

via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/4]

* 2001:DB8:ACAD::2/128

via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/4]

* 2001:DB8:ACAD::3/128

via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/4]

* 2001:DB8:ACAD::4/128

via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric 20
tag 0 LSP [2/4]

* 2001:DB8:ACAD:12::/64

```
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2  metric 20
tag 0 LSP [2/4]
```

```
* 2001:DB8:ACAD:23::/64
```

```
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2  metric 20
tag 0 LSP [2/4]
```

```
* 2001:DB8:ACAD:34::/64
```

```
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2  metric 20
tag 0 LSP [2/4]
```

```
* 2001:DB8:ACAD:45::/64
```

```
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2  metric 20
tag 0 LSP [2/4]
```

```
2001:DB8:ACAD:56::/64
```

```
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2  metric 20
tag 0 LSP [2/4]
```

```
R6#show isis * topology
```

Tag CORE:

IS-IS TID 0 paths to level-2 routers

System Id	Metric	Next-Hop	Interface	SNPA
R6	--			
R5 cc7f.7683.1400	10	R5	Gi0/0/0	

```
R6#show isis neighbors
```

Tag CORE:

System Id	Type	Interface	IP Address	State	Holdtime	Circuit Id
R5	L2	Gi0/0/0	10.0.56.1	UP	8	R5.01

Problems:

1. Lack of knowledge in redistribution

The hardest part of this lab wasn't configuring the protocols themselves since after doing it once or twice it becomes easy, and the protocols are straightforward. The thing that was the hardest was connecting them via redistribution and BGP. We fixed this by reading through cisco documentation to find the right metric numbers for redistribution and using the redistribute connected command on ipv6 and ipv4 for all protocols on both sides.

2. Lack of knowledge in IS-IS

IS-IS is an interior gateway protocol which was new to us. We had challenges understanding how the net-id worked, alongside configuring adjacencies. After looking at various examples of IS-IS configurations and cisco documentation, we were able to configure it properly.

3. Lack of knowledge in EIGRP

EIGRP, as mentioned in the background, is an advanced distance-vector protocol designed by Cisco as a proprietary protocol. This protocol was also new to us, which caused delays in configuring it internally. Like IS-IS, after spending time researching Cisco documentation and example configurations, we were able to set it up for both IPv4 and IPv6.

4. BGP Features

An extra part of our configuration was adding extra BGP features. These features influence BGP's path selection, and the two features we chose to implement were Weight and local preference. For weight, it's a number which is applied to a specific neighbor IP address, where the higher the number, the better. It is not advertised to neighbors, meaning it is locally significant, and it is used to prefer specific paths on a local router. The other feature we used was local preference, which is propagated within the Autonomous System, with the higher the number being better and the default set at 100. It is used to prefer exit points for outbound traffic. Both can be seen by doing `show ip bgp`.

5. Improper Topology

One major error when creating our original topology was having 4 separate ASN's for each of the border routers. While in theory this should have still worked, it complicated our configurations much more than necessary, and we fixed this by placing R3 and R4, the central routers in the same ASN instead of different ones.

6. *Non-Functional Routers*

Within our rack of routers, there were 2 routers which did not work because of corrupted iOS imaging and needed some debugging. We spent time attempting to debug

7. *Shared Routers*

The lab had us save our configurations on a notepad, then reconfiguring routers. The reason we had to do this was because other groups who we shared a lab rack with had consistently wiped out our router configs or changed them, which meant we spent a considerable amount of time rebooting routers.

8. *Restarting the lab*

During the middle of the lab, it was decided that we should restart the entire lab from scratch using our configs because there were numerous errors that we decided would take too long to debug manually. Especially for a lab of this small size, we knew that it would be faster to create a new bug-free config over debugging our incorrect configs.

Conclusion:

This lab covered a lot of new and valuable information about BGP. It was a great introduction into how the protocol works and interacts within its own ASN, and outside of its ASN, and it did a great job at showing the fundamental concept behind the benefit of BGP compared to other routing protocols.